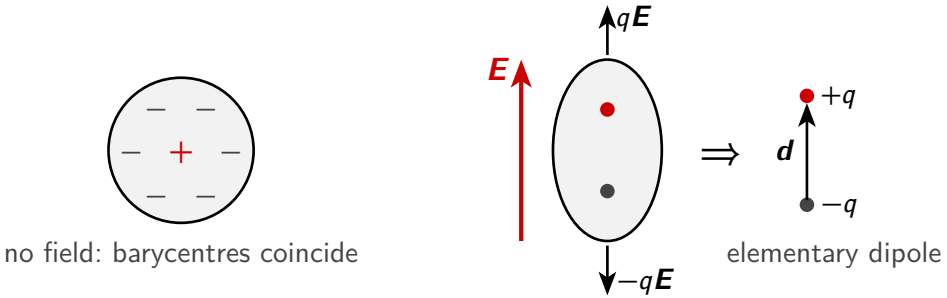
Chapter 3

Electromagnetism in matter

Electric polarisation - induced polarisation

Assume dielectrics: very low conductivity, $\sigma \sim 10^{-7} \div 10^{-8}$ S/m. **Non-polar** matter



no field: barycentres coincide

elementary dipole

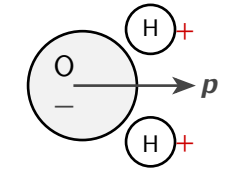
Without field, the barycentres of the positive and negative charges of an atom coincide

$\boldsymbol{E}$ pushes the positive charges along $\boldsymbol{E}$ ($q\boldsymbol{E}$) and the negative ones against it ($-q\boldsymbol{E}$): the barycentres separate, **induced** polarisation

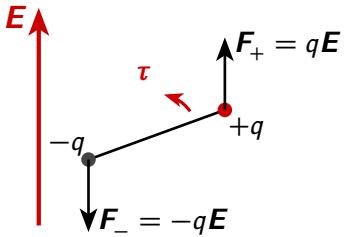
Elementary electric dipole: $\boldsymbol{p} = q\boldsymbol{d}$ [C m] $\boldsymbol{d}$ from $-q$ to $+q$

Electric polarisation - polarisation by orientation

Polar matter: the molecules have a **permanent** dipole moment



permanent dipole



Example H_2O: the electrons are closer to the oxygen, so O is slightly negative and the two H slightly positive

In a field $\boldsymbol{E}$: $\boldsymbol{F}_+ = q\boldsymbol{E}$ and $\boldsymbol{F}_- = -q\boldsymbol{E}$. The net force is zero, but the two forces form a **couple**

The torque turns the dipole towards $\boldsymbol{E}$: $\boldsymbol{\tau} = \boldsymbol{p} \times \boldsymbol{E}$ [N m]

thermal agitation opposes the alignment: the orientation is only partial

Electric polarisation - macroscopic description

Consider a point $\boldsymbol{x}$ and a small volume ΔV centred on it

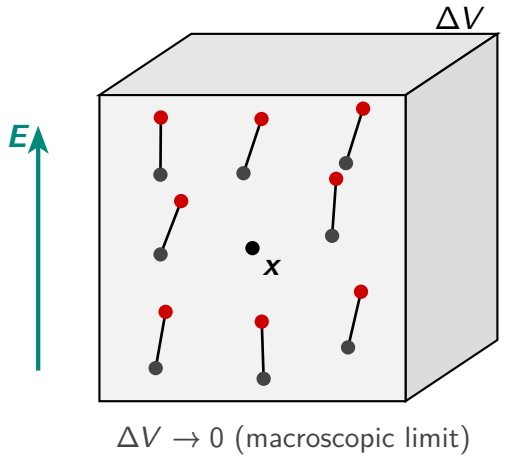
Resultant of the electric dipoles in ΔV: $\Delta \boldsymbol{p} = \sum_{i \in \Delta V} \boldsymbol{p}_i$

Polarisation field, as a macroscopic limit:

$$\boldsymbol{P}(\boldsymbol{x}) = \lim_{\Delta V \to 0} \frac{\Delta \boldsymbol{p}}{\Delta V} \quad [\text{C m/m}^3] = [\text{C/m}^2]$$

ΔV small on the macroscopic scale, but containing many dipoles

> $\boldsymbol{P}(\boldsymbol{x})$: the **local degree of polarisation** of matter



$\Delta V \to 0$ (macroscopic limit)

It can be proven that: $\nabla \cdot \boldsymbol{P} = -\rho_p$ (polarisation charge density)

Electric polarisation - Gauss law in matter

In matter, both free charges ρ and polarisation charges ρ_p are sources of $\boldsymbol{E}$

$$\nabla \cdot \boldsymbol{E} = \frac{\rho + \rho_p}{\varepsilon_0} \quad \Rightarrow \quad \nabla \cdot (\varepsilon_0 \boldsymbol{E}) = \rho - \nabla \cdot \boldsymbol{P}$$

Collect the divergences:

$$\nabla \cdot \left(\underbrace{\varepsilon_0 \boldsymbol{E} + \boldsymbol{P}}_{\boldsymbol{D}} \right) = \rho \quad \Rightarrow \quad \boxed{\nabla \cdot \boldsymbol{D} = \rho}$$

$\boldsymbol{D}$: **electric displacement** field [C/m^2]; only the **free** charges appear

Free space: $\boldsymbol{P} = 0 \Rightarrow \boldsymbol{D} = \varepsilon_0 \boldsymbol{E}$. In matter: $\boldsymbol{P} = \boldsymbol{P}(\boldsymbol{E}) \Rightarrow \boldsymbol{D} = \boldsymbol{D}(\boldsymbol{E})$

Electric polarisation - constitutive law

How does $\boldsymbol{P}$ depend on $\boldsymbol{E}$?

In general, through a **tensor**: $\boldsymbol{P} = \underline{\underline{\alpha}}\boldsymbol{E}$. Linear, **non-isotropic** material, e.g.

$$\underline{\underline{\alpha}} = \begin{pmatrix} 10 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \Rightarrow \quad P_x = 10E_x, \quad P_y = E_y, \quad P_z = E_z$$

off-diagonal terms: polarisation along one axis induced by field components along another

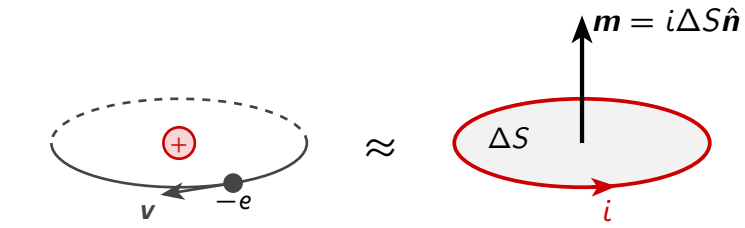
Linear, **isotropic** material: $\boldsymbol{P} = \varepsilon_0 \chi_e \boldsymbol{E}$, χ_e: electric susceptibility [−]

$$\boldsymbol{D} = \varepsilon_0 \boldsymbol{E} + \varepsilon_0 \chi_e \boldsymbol{E} = \varepsilon_0 \underbrace{(1 + \chi_e)}_{\varepsilon_r} \boldsymbol{E} \quad \Rightarrow \quad \boxed{\boldsymbol{D} = \varepsilon_0 \varepsilon_r \boldsymbol{E} = \varepsilon \boldsymbol{E}}$$

Relative permittivity ε_r: 1 (free space), ~ 1 (air), $2 \div 5$ (dielectrics), ~ 80 (water)

Magnetisation - elementary magnetic dipole

A current i around a small surface ΔS with normal $\hat{\boldsymbol{n}}$ is an elementary magnetic dipole



electron orbiting the nucleus ≈ elementary loop

Magnetic dipole moment: $\boldsymbol{m} = i\Delta S\hat{\boldsymbol{n}}$ [A m^2]

In an atom (classical picture), an electron orbiting the nucleus with period T is a small current loop, $i = e/T$

the current flows opposite to the motion of the electron, which is negative

Macroscopically, each atom or molecule behaves as an elementary loop with moment $\boldsymbol{m}$

the electron spin gives a magnetic moment too: it has no classical analogue, but it adds to $\boldsymbol{m}$ in the same way

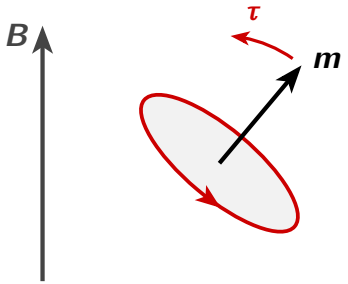
Magnetisation - orientation in a magnetic field

An elementary loop in a field $\boldsymbol{B}$

The Lorentz forces on the sides of the loop have zero resultant, but they form a **couple**

The torque turns $\boldsymbol{m}$ towards $\boldsymbol{B}$:

$$\boldsymbol{\tau} = \boldsymbol{m} \times \boldsymbol{B} \quad [\text{N m}]$$



same structure as $\boldsymbol{\tau} = \boldsymbol{p} \times \boldsymbol{E}$ for the electric dipole

The elementary loops orient along $\boldsymbol{B}$ and their effects add up: **magnetisation**

Magnetisation - macroscopic description

Consider a point $\boldsymbol{x}$ and a small volume ΔV centred on it

Resultant of the magnetic dipoles in ΔV: $\Delta \boldsymbol{m} = \sum_{i \in \Delta V} \boldsymbol{m}_i$

Magnetisation vector, as a macroscopic limit:

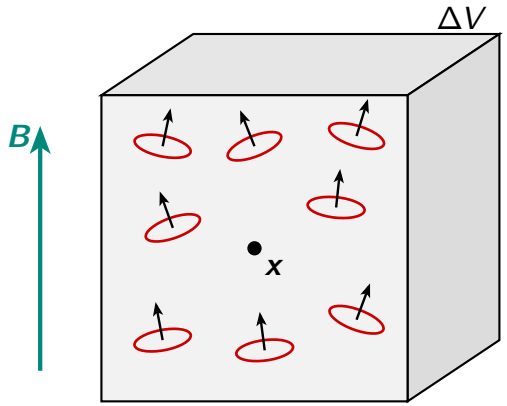
$$\boldsymbol{M}(\boldsymbol{x}) = \lim_{\Delta V \to 0} \frac{\Delta \boldsymbol{m}}{\Delta V} \quad [\text{A m}^2/\text{m}^3] = [\text{A/m}]$$

> $\boldsymbol{M}(\boldsymbol{x})$: the **local degree of magnetisation** of matter

It can be shown that:

$$\nabla \times \boldsymbol{M} = \boldsymbol{J}_m \quad \text{magnetisation current density}$$

$$\frac{\partial \boldsymbol{P}}{\partial t} = \boldsymbol{J}_p \quad \text{polarisation current density}$$



$\Delta V \to 0$ (macroscopic limit)

Magnetisation - Ampère-Maxwell in matter

In matter, magnetisation and polarisation currents add to the conduction current

From free space to matter:

$$\nabla \times \left(\frac{\boldsymbol{B}}{\mu_0} \right) = \boldsymbol{J} + \frac{\partial(\varepsilon_0 \boldsymbol{E})}{\partial t} + \boldsymbol{J}_m + \boldsymbol{J}_p = \boldsymbol{J} + \frac{\partial(\varepsilon_0 \boldsymbol{E})}{\partial t} + \nabla \times \boldsymbol{M} + \frac{\partial \boldsymbol{P}}{\partial t}$$

Collect the curls on the left and the time derivatives on the right:

$$\nabla \times \left(\underbrace{\frac{\boldsymbol{B}}{\mu_0} - \boldsymbol{M}}_{\boldsymbol{H}} \right) = \boldsymbol{J} + \frac{\partial}{\partial t} \left(\underbrace{\varepsilon_0 \boldsymbol{E} + \boldsymbol{P}}_{\boldsymbol{D}} \right)$$

$$\nabla \times \boldsymbol{H} = \boldsymbol{J} + \frac{\partial \boldsymbol{D}}{\partial t} \quad \boldsymbol{H} = \frac{\boldsymbol{B}}{\mu_0} - \boldsymbol{M} \ \ [\mathrm{A/m}]$$

Magnetisation - constitutive law

We found $\boldsymbol{H} = \boldsymbol{B}/\mu_0 - \boldsymbol{M}$; solving for $\boldsymbol{B}$:

$$\boldsymbol{B} = \mu_0(\boldsymbol{H} + \boldsymbol{M})$$

In general $\boldsymbol{M}$ depends on $\boldsymbol{H}$: $\boldsymbol{M} = \boldsymbol{M}(\boldsymbol{H}) \Rightarrow \boldsymbol{B} = \boldsymbol{B}(\boldsymbol{H})$

Assume linear, isotropic materials: $\boldsymbol{M} = \chi_m \boldsymbol{H}$, χ_m: magnetic susceptibility [–]

$$\boldsymbol{B} = \mu_0(\boldsymbol{H} + \chi_m \boldsymbol{H}) = \mu_0 \underbrace{(1 + \chi_m)}_{\mu_r} \boldsymbol{H} \quad \Rightarrow \quad \boxed{\boldsymbol{B} = \mu_0 \mu_r \boldsymbol{H} = \mu \boldsymbol{H}}$$

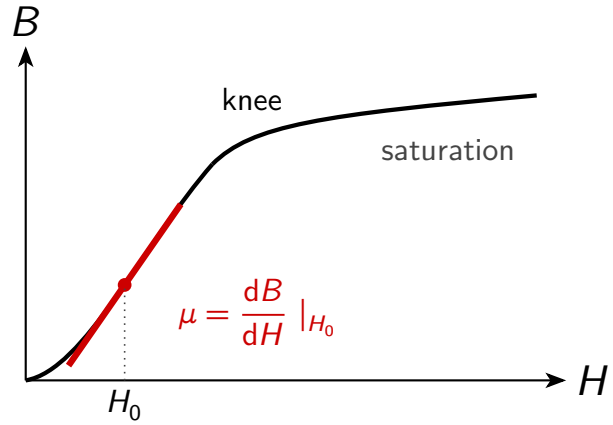
Magnetisation - relative permeability

material	μ_r
free space	1
air	≈ 1
ferromagnetic (soft)	$10^3 \div 10^4$

Ferromagnetic materials are **non-linear**: the B-H curve bends at the **knee** and saturates

μ comes from the curve **linearised** around a working point H_0:

$$\mu = \frac{\mathrm{d}B}{\mathrm{d}H}\,|_{H_0}, \quad \mu_r = \frac{\mu}{\mu_0}$$



Constitutive laws - local Ohm's law

Constitutive laws so far (linear, isotropic materials):

$$\boldsymbol{D} = \varepsilon_0 \varepsilon_r \boldsymbol{E}$$
$$\boldsymbol{B} = \mu_0 \mu_r \boldsymbol{H}$$
$$\boldsymbol{J} = \sigma(\boldsymbol{E} + \boldsymbol{E}_i)$$

Third law, for conductors: the **local Ohm's law** (in red), with conductivity σ [S/m]

$\boldsymbol{E}$: the electric field of Maxwell's equations, **including** the field induced by $\partial \boldsymbol{B}/\partial t$ (Faraday)

conservative only in (electro)statics

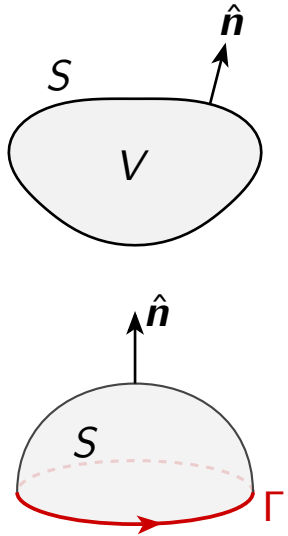
$\boldsymbol{E}_i$: **impressed** (electromotive) fields, of **non-electromagnetic** origin: electrochemical fields, temperature or electron density gradients, ...

not part of $\boldsymbol{E}$ and not described by Maxwell's equations: the sources ("generators") of the problem

Maxwell's equations in matter

Local (differential) form $\Leftrightarrow$ global (integral) form, via the divergence and Stokes theorems

	differential	**integral**
Gauss	$\nabla \cdot \boldsymbol{D} = \rho$	$\oint_S \boldsymbol{D} \cdot \hat{\boldsymbol{n}} \, \mathrm{d}S = Q$
no magnetic charges	$\nabla \cdot \boldsymbol{B} = 0$	$\oint_S \boldsymbol{B} \cdot \hat{\boldsymbol{n}} \, \mathrm{d}S = 0$
Faraday-Neumann-Lenz	$\nabla \times \boldsymbol{E} = -\dfrac{\partial \boldsymbol{B}}{\partial t}$	$\oint_\Gamma \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{l} = -\dfrac{\mathrm{d}\Phi}{\mathrm{d}t}$
Ampère-Maxwell	$\nabla \times \boldsymbol{H} = \boldsymbol{J} + \dfrac{\partial \boldsymbol{D}}{\partial t}$	$\oint_\Gamma \boldsymbol{H} \cdot \mathrm{d}\boldsymbol{l} = i + i_d$



Electromagnetism in matter - the complete model

Hp. linear and isotropic materials: the full set of equations

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}$$
$$\nabla \times \boldsymbol{H} = \boldsymbol{J} + \frac{\partial \boldsymbol{D}}{\partial t}$$
$$\nabla \cdot \boldsymbol{B} = 0, \quad \nabla \cdot \boldsymbol{D} = \rho$$

$$\boldsymbol{D} = \varepsilon \boldsymbol{E}$$
$$\boldsymbol{B} = \mu \boldsymbol{H}$$
$$\boldsymbol{J} = \sigma(\boldsymbol{E} + \boldsymbol{E}_i)$$

Continuity $\frac{\partial \rho}{\partial t} + \nabla \cdot \boldsymbol{J} = 0$ still follows from Ampère-Maxwell and Gauss

Unknowns: $\boldsymbol{E}, \boldsymbol{D}, \boldsymbol{H}, \boldsymbol{B}, \boldsymbol{J}$; sources: the impressed fields $\boldsymbol{E}_i$

Next: the **energy** balance (Poynting) and the **uniqueness** of the solution